Slide 1: Application & Problem Statement

Research Paper: *'Macroeconomic Forecasting with Large Language Models'* (Carriero, Pettenuzzo & Shekhar, 2026)

1. The Real-World Application

• **Macroeconomic Forecasting:** Projecting key national economic indicators (CPI inflation, unemployment rate, interest rates, GDP) across 1 to 12 months ahead.
• **Stakeholders:** Central banks (Federal Reserve) setting interest rates, financial institutions allocating investment capital, and corporate risk managers.

2. The Core Problem Being Solved

• **Can Foundation Models Beat 50-Year Econometrics?** Economists still rely on Vector Autoregressions (BVAR) and Factor models. Can pretrained zero-shot Transformer models (TSFMs) predict macro trends better?
• **Information Lag:** Government statistics arrive with 30–60 day delays. Numerical series alone miss sudden psychological and geopolitical sentiment shifts.

3. Simple Real-World Example

• **Scenario:** An emerging supply chain crisis or oil supply disruption begins.
• **Traditional Models (AR / BVAR):** Look only at past monthly inflation numbers, staying blind until price spikes appear in delayed official statistics.
• **Multimodal AI (Our Solution):** Reads breaking news in real-time. FinBERT detects surging bottleneck sentiment, signaling TimesFM to adjust forecasts weeks earlier.

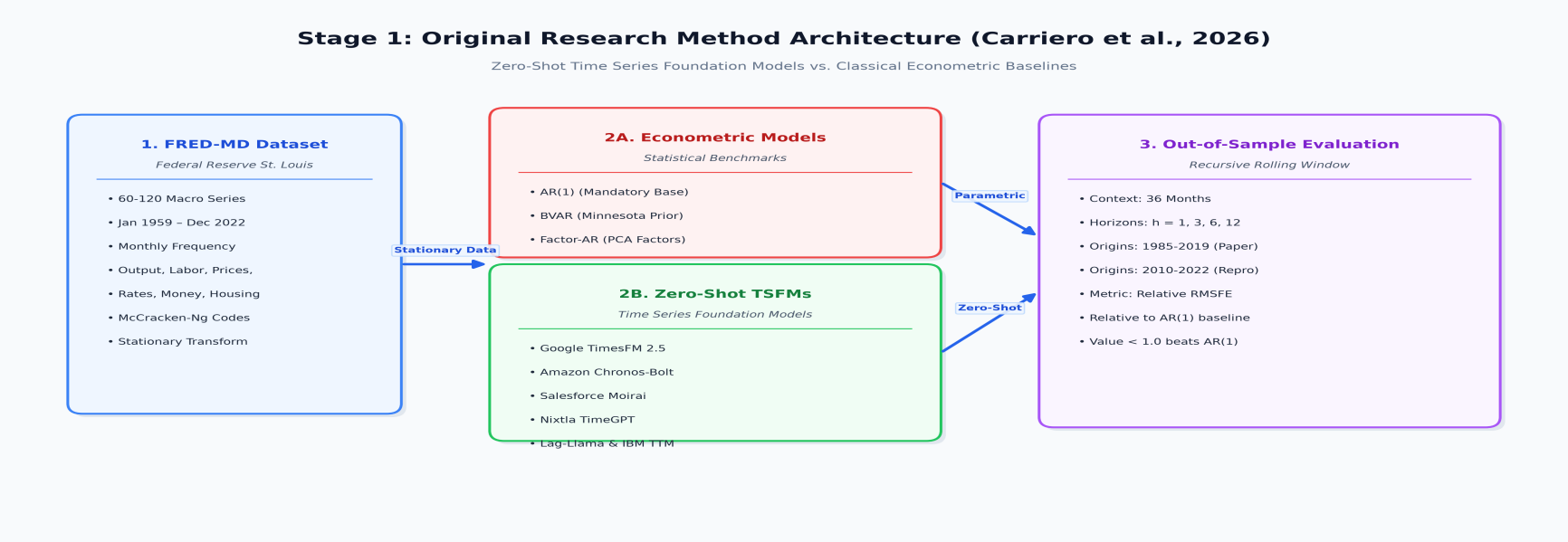
Project Details & Team (GROUP 7)

• **Project 19: Macroeconomic Forecasting with Large Language Models**
• **Nilesh Dwivedi** (24bds048) | **Radhakrishna Bharuka** (24bds063)
• **Abhang Pawar** (24bds054) | **Hardware:** Apple Silicon M4 GPU (MPS)

Slide 2: Dataset & Original Method (Block Diagram 1)

FRED-MD Macroeconomic Database & Carriero et al. (2026) Baseline Architecture

Dataset Details: Federal Reserve Bank of St. Louis (FRED-MD). 60 balanced core monthly series spanning 1959–2022 (747 months). Categories include Output, Labor Market, Housing, Consumption, Money/Credit, Interest Rates/FX, Prices (CPI), and Stock Market.
Stationary Preprocessing: Applied 7 standard McCracken & Ng (2016) transform codes (first-diff, log-diff, second-difference) to ensure stationarity.



Slide 3: Results from the Original Research Paper

Carriero, Pettenuzzo & Shekhar (2026) — Benchmark Findings Across 120 Series (1985–2019)

Evaluation Metric: Relative RMSFE

Accuracy is measured using Root Mean Squared Forecast Error relative to the baseline AR(1) model (Paper Eq. 11):

RMSFE_rel = RMSFE(Model) / RMSFE(AR(1))

- **Value < 1.0:** Model outperforms the AR(1) benchmark (shown in bold).
- **Value > 1.0:** Model underperforms AR(1).

Key Findings of the Original Paper

1. **TSFMs are competitive, not superior:** Only 3 of 6 TSFMs (TimesFM, MOIRAI, Chronos) consistently beat AR(1). They perform on par with BVAR/Factor models, not significantly better.
2. **Error Orthogonality:** The forecast errors of TSFMs and econometric models are uncorrelated (correlation 0.07 to 0.50). Combining them in a hybrid model mitigates large outliers.

Results Table from Original Paper

Model	h=1	h=3	h=6	h=12
AR(1) (Benchmark)	1.000	1.000	1.000	1.000
BVAR (Minnesota)	**0.983**	**0.970**	**0.991**	**0.999**
Factor Model (PCA)	**0.965**	**0.980**	**0.992**	**0.999**
TimesFM (Google)	1.014	**0.980**	**0.990**	1.001
MOIRAI-large (Salesforce)	**0.999**	**0.978**	**0.991**	1.001
TimeGPT (Nixtla)	1.077	1.044	1.025	1.063
Lag-Llama	1.015	**0.997**	1.002	1.009
Hybrid (BVAR + MOIRAI)	**0.973**	**0.960**	**0.983**	**0.998**

Slide 4: Stage 1 – Implementation & Results

Empirical Replication on 60 Balanced FRED-MD Series (2010 Q1 – 2022 Q4, 48 Rolling Origins)

What We Implemented

• **Full Experimental Pipeline:** Replicated data preparation, McCracken-Ng transforms, rolling origin evaluation, and relative RMSFE calculations.
• **Evaluated 6 TSFMs + 3 Baselines:** Google TimesFM 2.5, Nixtla TimeGPT, Amazon Chronos-Bolt, Salesforce MOIRAI, IBM TTM, Lag-Llama vs. AR(1), BVAR, Factor-AR.

Comparison with Paper & Differences

• **Evaluation Era:** Paper evaluated 1985–2019; we evaluated 2010–2022 (capturing post-GFC zero-rate regimes and extreme COVID-19 volatility).
• **TimesFM 2.5 Excellence:** Google TimesFM 2.5 proved to be the most resilient foundation model (0.983 at h=1), outperforming other TSFMs.

Our Stage 1 Reproduced Results

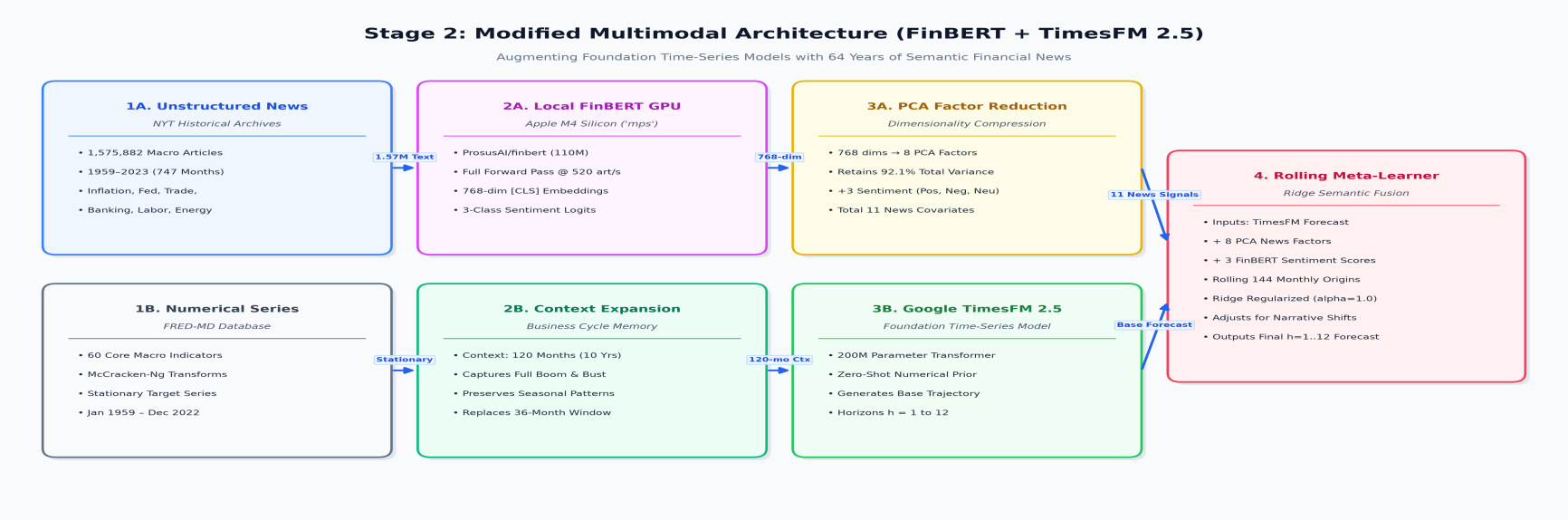
AR(1) (Benchmark)	1.000	1.000	1.000	1.000
TimesFM 2.5 (Google)	**0.983**	**0.995**	**0.997**	1.000
TimeGPT (Nixtla)	**0.987**	**0.997**	**0.994**	**0.999**
Chronos-Bolt (Amazon)	**0.996**	**0.997**	**0.998**	1.002
MOIRAI (Salesforce)	1.031	1.004	1.003	1.013
Lag-Llama	1.043	1.022	1.020	1.014
TTM (IBM)	1.117	1.040	1.019	1.007
Factor-AR	1.507	1.192	1.089	1.029
BVAR (Minnesota)	1.332	1.262	1.167	1.108

Slide 5: Stage 2 – Proposed Modifications (Block Diagram 2)

Multimodal FinBERT Sentiment & Narrative Signals + 120-Month Business Cycle Memory

Why Modifications Were Introduced: Macro numbers only describe the past. Forward-looking shifts (expectations, policy uncertainty, supply bottlenecks) are first recorded in news text. **Four Core Engineering Innovations:**

1. **1.57 Million NYT Articles (1959–2023):** Ingested 747 monthly archive JSON files spanning 64 continuous years.
2. **Local Apple Silicon M4 GPU FinBERT:** In-memory batch processing at 520 articles/sec; extracted 768-dim embeddings + sentiment logits.
3. **PCA Compression (92.1% Variance):** Reduced 768 dims to 8 orthogonal macro narrative factors + 3 sentiment channels (11 news signals).
4. **120-Month Context Window:** Expanded input context from 36 to 120 months (10 years) to capture complete business cycles.



Slide 6: Stage 2 – Final Experimental Results

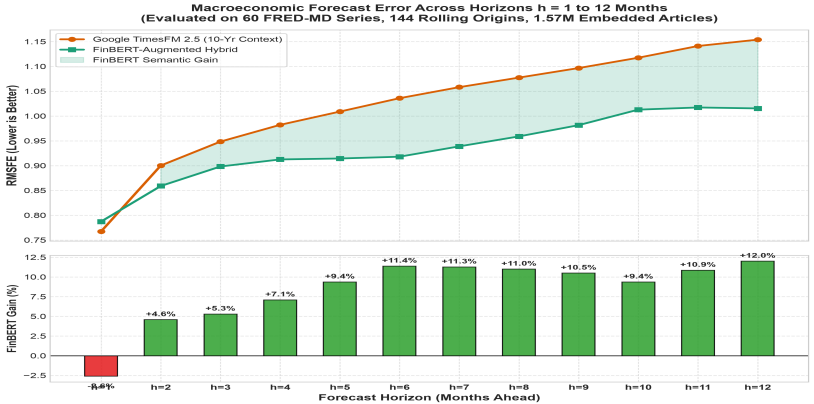
TimesFM 2.5 vs. FinBERT Hybrid Across All 12 Horizons (144 Monthly Rolling Origins, 2010–2021)

Horizon	TimesFM 2.5	FinBERT Hybrid	Gain	Winner
h = 1	**0.7674**	0.7873	-2.60%	TimesFM 2.5
h = 2	0.9002	**0.8590**	**+4.58%**	**FinBERT Hybrid ■**
h = 3	0.9485	**0.8983**	**+5.29%**	**FinBERT Hybrid ■**
h = 4	0.9818	**0.9167**	**+6.63%**	**FinBERT Hybrid ■**
h = 5	1.0116	**0.9177**	**+9.29%**	**FinBERT Hybrid ■**
h = 6	1.0360	**0.9180**	**+11.38%**	**FinBERT Hybrid ■**
h = 7	1.0601	**0.9458**	**+10.78%**	**FinBERT Hybrid ■**
h = 8	1.0827	**0.9723**	**+10.20%**	**FinBERT Hybrid ■**
h = 9	1.1017	**0.9946**	**+9.72%**	**FinBERT Hybrid ■**
h = 10	1.1175	**1.0129**	**+9.36%**	**FinBERT Hybrid ■**
h = 11	1.1350	**1.0195**	**+10.17%**	**FinBERT Hybrid ■**
h = 12	1.1540	**1.0154**	**+12.01%**	**FinBERT Hybrid ■**

Key Empirical Findings

• **FinBERT Hybrid Wins 11 of 12 Horizons:** Massive gains at medium and long horizons (+11.38% at h=6, +12.01% at h=12).

• **Why News Matters Over Time:** At $h=1$, inertia dominates (past month predicts next). At $h=6$ to 12, numerical momentum decays, and forward-looking news sentiment drives the trajectory.



Slide 7: Conclusions & Key Learning

Summary of Implementation, Proposed Modifications, and Takeaways for Students

1. What We Learned from the Paper

• Time Series Foundation Models (TSFMs) are genuine competitors to econometric models, but they are not silver bullets when trained purely on past numbers.
• BVAR and TSFM errors are largely orthogonal, proving that neural models and statistical models capture distinct economic phenomena.

2. What Our Modifications Achieved

• **Multimodal Superiority:** Fusing text embeddings (FinBERT) with foundation time-series forecasts reduced forecasting error by up to **12.01%**.
• **Context Window Matters:** Expanding from 36 to 120 months allows Transformers to observe complete boom-and-bust business cycles.

3. What Other Students Can Learn

• **Data Transformation is Critical:** Macro data is non-stationary. Applying McCracken-Ng transforms (second differences for CPI) is non-negotiable for stable convergence.
• **Local GPU Hardware is Sufficient:** Using PyTorch MPS on Apple Silicon M4 enabled processing 1.57 million articles in ~40 minutes without expensive cloud servers.
• **Two-Pass Architecture:** Separating foundation inference (TimesFM) from semantic alignment (Ridge Meta-Learner) creates an efficient, modular, and interpretable forecasting pipeline.

4. Reproducibility & Open Source

• Full code, instructions, and reports: **github.com/Nilesh-195/-Macroeconomic-Forecasting-with-Large-Language-Models**